Promising Less to Be Wrong Less Often: Four Renunciation Mechanisms as Alternative Actions in Crop Mapping

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Abstract. An uncertain crop map can shrink its promised catalogue, abstain, return plausible labels, or back off to a coarser agronomic category. Prior work treats these four actions as separate lines with separate optima and evaluations. Within our bounded search, we found no comparison that puts them in one decision problem and asks what follows without choosing an outcome price. We state the estimand and population first, fix every operating point on training and validation data, and retain the complete six-coordinate loss space. The data identify outcome-frequency vectors, uniform dominance, and linear switching boundaries; stakeholder preference requires an external loss point. In exploratory PASTIS-R evaluation, none of 75 within-predictor action pairs was certified as dominant: 71 crossed and 4 remained unresolved. A unique winner was identified in 3913 of 6600 model–loss evaluations on the declared lattice; the precise baseline, abstention, adaptive sets, and taxonomic back-off each won somewhere. These counts are not simplex volume, and no admissible confirmatory benchmark was found in our bounded search. Retrospectively, a twelve-class deployed catalogue delivered on 15265 of 16640 parcels, and none of thirty frozen alternative system configurations dominated its profile. Thus evidence determines switching regions, not one best way to promise less.

Keywords: Crop mapping · Selective classification · Set-valued prediction · Decision theory · Satellite image time series

1 Introduction

A parcel-level crop map is a promise. For every field in the scene it names a crop, and somebody downstream acts on that name: an inspector schedules a visit, an agronomist plans an input, an agency settles a subsidy. When the classifier is unsure, answering anyway transfers the uncertainty to that person without telling them.

There is more than one way to say less. The map can *shrink the catalogue* it is willing to promise and stop offering the classes it cannot separate; it can *abstain* on a parcel and hand it to a human; it can return a *set of candidate labels* and let the reader pick; or it can *fall back* to a coarser agronomic category that it can defend. Each of these buys accuracy with a different currency, and each one moves the cost somewhere else.

Promising less to be wrong less often is not a new idea. The error–reject trade-off is due to Chow [3], and Ha later generalized it to selecting a *subset* of classes—together with the optimality criterion that governs it [10]. Recent work gives efficient algorithms for the subset of maximum expected utility [14] and a systematization of the available formulations [4]. In crop mapping specifically, catalogue size has already been measured as a design variable [7], and the closest work to ours rejects pixels by uncertainty on the same benchmark and on three architectures also represented in our original candidate bank, making the trade-off between precision and recall explicit [16].

What we did not find is a comparison of these mechanisms *against each other*. Each line of work optimizes its own device and reports its own curve. Within the search scope described in Section 2, we found no work that puts catalogue shrinking, abstention, set-valued answers, and taxonomic back-off in one decision problem, preserves the complete loss space, and asks which conclusions are invariant to every admissible outcome price.

Contribution. We state the comparison and the conditions under which it can be believed. Concretely: (i) we cast the four mechanisms as alternative *actions* of the same decision problem and represent each by its complete outcome-frequency vector; (ii) we fix the estimand, the population, and the unit of analysis *before* computing anything, and we require the operating point of every mechanism to come from training and validation data alone; (iii) we derive uniform dominance and switching boundaries over the full normalized loss space, rather than choosing a favourable point; and (iv) we make the distribution of outcomes across classes a reported quantity rather than a side effect, because a mechanism that improves an average by silently dropping rare classes has not improved the same map.

What this paper does not claim. We do not infer a stakeholder's preferences from model outputs and do not choose a loss point on their behalf. The evidence can identify uniform dominance and the boundary at which one action becomes preferable to another; it cannot say where a particular inspector, adviser, or producer lies in that space. Any later external loss point is a declared sensitivity case, not a hidden input to the primary analysis. The recorded candidate search closed without an admissible confirmatory dataset, and its dataset-specific query log was not versioned; the resulting absence claim is therefore limited to the documented candidate set rather than the whole literature. The two available parcel datasets remain exploratory. The conformal and hierarchical diagnostics reported below do not change that role and are not a completed comparison of the four actions.

2 Related Work

We organize this section by *limitation* rather than by application: for each line of work we state what it establishes and where it stops, because the stopping point is what defines the work that is left.

The renunciation framework already exists. The error–reject trade-off and its optimal rule are due to Chow [3]; that formulation is the degenerate case in which the only alternative to answering is silence. Ha generalized it to class-selective rejection, showing that selecting a subset of classes is the optimal compromise between error and the mean number of classes returned [10]; the treatment is decision-theoretic and assumes the posteriors are known. Ha's class-selective rule is an implemented parameterized literature family, not a seventh empirical arm: no cost ratio has been selected. Adding one would require a prospective amendment and a new power calculation. Mortier et al. give algorithms for the Bayes-optimal subset of maximum expected utility, with a utility that is a function of set size [14], and Chzhen et al. unify the existing set-valued formulations and their trade-offs [4], at infinite sample and under a plug-in principle. The framework is therefore not our contribution, and we say so on the first page. What these treatments share is that the price of an answer is a function of *how large* the answer is. They do not compare the four delivery actions through the same complete outcome-specific loss space.

Learned abstention is developed, and out of our scope by choice. Deep selective classification supplies risk guarantees [5], an architecture with an integrated reject head [6], a wagering-style loss [12], and an AUC-based selection criterion [15]. All of them abstain *per instance*; none emits a set or falls back to a coarser class. We do not reimplement them and we do not benchmark against them: our question is how abstention compares with the other three actions across a common loss space, not whether a better abstention rule exists.

Renunciation has reached Earth observation, and not through agriculture. The closest work to ours rejects pixels by an uncertainty threshold on PASTIS, using three architectures represented in our original candidate bank, and makes the trade-off explicit: not answering lowers recall and can raise precision when the rejected pixel would have been misclassified [16]. That is the risk–coverage compromise, performed and acknowledged. What it does not do is place the decision in the selective-classification frame, compare alternative actions, or characterize the loss-space boundaries at which the preferred action changes. Elsewhere in Earth observation, SHRUG-FM reports abstention with interpretable thresholds [9], on burn scars, floods, and landslides rather than on any agricultural task. Rejection methods have been applied to vegetation mapping [8], on airborne hyperspectral data rather than per-parcel satellite time series. Open-set crop recognition rejects the *unknown* through outlier exposure [2], which is a different question from distributing a degree of promise among known classes.

Catalogue size is already a measured design variable. Ghassemi et al. treat the number of classes as a design variable and measure it: starting from 52 LUCAS classes, a 26-class scheme balances accuracy against detail [7]. This is the direct precedent for reading catalogue shrinking as an operating point, and it forecloses any claim that we are the first to do so. There the setting is land cover with Random Forest; here it is one mechanism among four, represented in a common outcome vocabulary over the complete admissible loss space.

Taxonomic back-off already has a crop-mapping precedent. Multi-scale label hierarchies improve crop mapping from satellite image time series [19]. That work both predicts at three granularities and allows the output granularity to follow classification confidence; it is therefore a direct precedent for confidence-driven taxonomic back-off, not merely for hierarchical training. Hierarchical fusion also combines satellite, rotational, and contextual signals [1], but for information fusion rather than variable-granularity delivery. We do not claim the back-off mechanism as new. Hierarchical set-valued prediction has also been formalized with internal nodes as the usual valid sets [13]. The open comparison is whether this action is uniformly preferable to another action, and where their preference boundary lies otherwise.

Distributional effects are documented outside our domain. Selective classification can *magnify* disparities between groups: abstaining more can make outcomes worse for some of them [11]. The evidence is on demographic groups in vision and language, not on agronomic classes or parcels. We take it as the reason to report distribution across classes rather than as evidence about crops, and we note that the result runs opposite to the intuition that declining to answer is a neutral act.

Novelty, stated with its universe. We found no work that compares catalogue shrinking, per-parcel abstention, set-valued prediction, and taxonomic back-off *as alternative actions of the same decision problem*, reports their uniform-dominance order over the complete outcome-specific loss space, and accounts for how the underlying outcomes are distributed across classes. This statement is bounded by our search: arXiv, OpenAlex, and Semantic Scholar, 61 queries plus 6 manual ones, covering 2019–2026 together with the classics retrieved by name, in English only. That procedure indexes remote-sensing proceedings without DOIs poorly, and it is written in the vocabulary of selective classification—a limitation we observed in practice, since the closest work to ours [16] uses none of those terms and was found by another route.

3 Problem Statement and Estimand

3.1 Four Ways to Promise Less

Let $\mathcal{Y}$ be the class catalogue, let x_i be the observations for parcel i, and let $y_i \in \mathcal{Y}$ be its reference label. A conventional classifier commits to a map $x_i \mapsto \hat{y}_i \in \mathcal{Y}$: it must name exactly one class for every parcel, whatever its posterior looks like. The mechanisms compared here all relax that commitment in the same formal way. Each replaces the point prediction with a subset of the catalogue,

$$C_m(x_i) \subseteq \mathcal{Y}, \qquad C_m(x_i) = \emptyset \text{ permitted}, \tag{1}$$

and they differ only in which subsets they may emit, and in what fixes the choice.

Catalogue shrinking. A subcatalogue $\mathcal{S} \subset \mathcal{Y}$ is fixed before any parcel is seen. A parcel whose unrestricted posterior maximizer lies inside $\mathcal{S}$ receives the restricted maximizer; a parcel whose maximizer lies outside receives nothing. The rule reads the posterior and never the reference label—keying delivery on the truth would make the mechanism unimplementable at inference time, and an earlier implementation in this line of work did exactly that. The decision is per class, and it is taken once.

Per-parcel abstention. The mechanism emits a singleton or the empty set, according to whether a confidence score clears an operating point. The decision is per parcel, and it is taken as many times as there are parcels.

Set-valued prediction. The mechanism emits a non-empty subset whose size varies with the difficulty of the parcel, calibrated so that a stated guarantee holds over the population. Every parcel receives an answer; what varies is how much the answer commits to.

Taxonomic back-off. The mechanism emits the set of leaves under a node of a class hierarchy. Formally this is a subset like any other; what distinguishes it is that the admissible subsets are those an agronomist can name, so the answer degrades to a coarser but still communicable category instead of to a list.

Two degenerate elements of the same space bound it: full-catalogue singleton prediction, which renounces nothing, and universal non-delivery, for which $C \equiv \emptyset$. Neither is one of the mechanisms, because neither chooses what to give up. They are not on the same footing, however, and Section 4 keeps them apart: the unmodified singleton is evaluated, as the reference against which every renunciation is priced, whereas universal non-delivery only fixes the price of silence and is never run. Writing all of these in one action space is this paper's first move, and it is what makes the question askable at all: as long as legend design, thresholding, conformal calibration, and hierarchical prediction are treated as separate techniques, there is no object on which to ask which of them one should prefer.

3.2 The Estimand

Fix a dataset d with its region, campaign, split, and predictor, and a mechanism m operating at τ_m. Each eligible held-out parcel falls into exactly one cell of a closed outcome account: correct precise delivery, incorrect precise delivery, non-delivery, a set that contains or excludes the truth, or a coarse answer that contains or excludes the truth. Correct precise delivery is the zero-loss reference R_0; write the remaining cells as $L_1, \dots, L_6$. The empirical quantity is their frequency vector,

$$q_{d,m,j} \;=\; \frac{1}{N_d} \sum_{i=1}^{N_d} \mathbf{1}\{O_i(m) = L_j\}\,, \qquad j = 1, \dots, 6, \tag{2}$$

where N_d counts *all* eligible parcels of that block. For any finite non-negative loss vector, common positive scale is irrelevant to the ordering, so we normalize it to

$$\boldsymbol{\ell} \in \Delta^5 = \left\{ \boldsymbol{\ell} \in \mathbb{R}^6_+ : \sum_{j=1}^{6} \ell_j = 1 \right\}, \qquad R_{d,m}(\boldsymbol{\ell}) = \boldsymbol{q}_{d,m}^{\mathsf{T}} \boldsymbol{\ell}. \tag{3}$$

The data determine $\boldsymbol{q}$, not $\boldsymbol{\ell}$. We therefore report the full linear decision geometry instead of inserting a point. Arm m uniformly dominates arm n exactly when $q_{m,j} \le q_{n,j}$ for every cost-bearing cell and at least one inequality is strict. Otherwise, their switching boundary is the hyperplane $(\boldsymbol{q}_m - \boldsymbol{q}_n)^{\mathsf{T}} \boldsymbol{\ell} = 0$. A scalar switching price exists only inside a declared section that fixes the other coordinates. A two-dimensional picture is consequently a sensitivity view, not the complete five-dimensional space.

Cardinality is not the loss. Expected set size prices two sets of equal size identically even when one excludes the truth, and it assigns zero size to non-delivery. We retain it as a descriptor but never use it as the common objective. Nor do we infer a loss point from the model outputs: that would turn a modeling preference into a data result. A later externally sourced point may be overlaid as a non-confirmatory sensitivity case, with its provenance stated.

The population includes the parcels that receive nothing. N_d ranges over every eligible parcel in the block, including those the mechanism declines to answer and those whose class the shrunken catalogue no longer contains. Restricting the denominator to delivered parcels would price renunciation at zero and would score each mechanism on a population of its own choosing, so that a mechanism could improve its reported quality by answering less.

The operating point is fixed before the held-out block is touched. τ_m is derived entirely from training and validation. The held-out block estimates the realized outcome and nothing else. Re-equalizing loss, coverage, or delivery rate on the held-out block is prohibited, including as a post-hoc alignment intended to make mechanisms comparable: it is the leak that two audits of an earlier version of this work found, first in the threshold and then in its target rate.

3.3 Scope, Dependence, and What Is Not Claimed

Inference is *conditional on each dataset*, with its region, campaign, split, and predictor. The observational unit is the parcel. Parcels are not independent: they arrive in image patches that share acquisition, geography, and annotation, so the dependence cluster is the patch. The inferential bootstrap therefore resamples whole patches within spatial blocks and never parcels one by one. A separate paired interval over spatial-block means is reported beside it to expose between-block variation; because those folds share training data, it is a sensitivity view rather than evidence from independent territories. Below three unique paired units for either view, that view is reported descriptively — without interval, without p-value, and without multiplicity correction — because at that size the interval would describe the resampling scheme and not the data.

No quantity is pooled across datasets. Each dataset is reported on its own, and the second one serves as a replication attempt rather than as an additional sample of the first. The number of spatial blocks is a sensitivity parameter for the spatial structure of the estimate, not a count of replicates and not a lever on statistical power: partitioning the same territory more finely makes the resulting folds share more of their training data, so it produces the same evidence counted more often.

What this design gives up is stated here rather than in a limitations paragraph, because it constrains every claim that follows: **we do not claim transport**. Nothing reported for one region or campaign is asserted to hold in another. A conditional estimand is what the available partitions support; a marginal one would require a sample of regions that this work does not have, and the two were mixed in the design of an earlier version until an audit separated them.

The intended deployment context is Mexico, but our bounded search identified no public Mexican benchmark with parcel-level crop labels. The datasets available to this study therefore do not validate the deployment geography. This is a limitation of the evidence, not proof that such data do not exist, and no result below is generalized from the evaluated regions to Mexico.

4 Four Mechanisms as Alternative Actions

The comparison contains four mechanisms but six experimental arms. This distinction matters. There is one unmodified singleton baseline; catalogue shrinking has two arms, because the retained catalogue is ranked once by validation F-score and once by training support; and abstention, set-valued prediction, and taxonomic back-off contribute one arm each. Universal non-delivery is a boundary action used to identify the non-delivery cell, not a seventh experimental arm. Thus the count is

$$\text{baseline} + \text{two catalogue rules} + \text{three other mechanisms} = \text{six arms},$$

while the scientific question remains a comparison of four ways to renounce commitment. This also prevents a bookkeeping ambiguity in Section 3.1: the two boundary actions define the space, but they are not the two extra empirical arms.

4.1 A Common Output Type

Every arm returns the Boolean membership vector of $C_m(x_i)$ over the fixed catalogue. The representation contains no reference label, loss, or delivered-only filter. A row with one true entry is a precise delivery, a row with several true entries is an ambiguous or coarsened delivery, and an all-false row is non-delivery. Keeping the empty set inside the output type is important: it ensures that an abstained parcel remains in the population rather than disappearing before it can be counted.

The baseline emits the singleton containing the posterior maximizer. Both catalogue-shrinking arms apply the same operator to a different predeclared subcatalogue. If the unrestricted posterior maximizer belongs to that subcatalogue, it is emitted; otherwise the output is empty. The operator does not inspect the reference label. The two arms therefore differ only in the training-side rule that orders classes for withdrawal, not in how a test parcel is handled.

Per-parcel abstention emits the posterior maximizer when its confidence clears a fixed threshold and the empty set otherwise. Set-valued prediction sorts labels by posterior probability and emits the smallest non-empty prefix reaching a calibrated cumulative mass. The constructor applies that mass but never estimates it: calibration belongs entirely to training and validation, and any coverage guarantee depends on that preceding calibration rather than on the constructor alone.

The set-valued arm uses deterministic, non-randomized split-conformal adaptive prediction sets. For each calibration parcel, the score is the cumulative posterior mass through the true label under a stable probability ranking. The calibrated mass is selected by the exact finite-sample order-statistic rank $\lceil (n_{\mathrm{cal}} + 1)(1-\alpha)\rceil$, not by passing a rank fraction to an interpolating quantile API. If that rank lies beyond the calibration sample, the arm emits the complete catalogue rather than substituting the largest observed score. We choose the deterministic convention to avoid introducing an auxiliary randomization and seed; its possible conservatism is reported through observed marginal containment and set size. Held-out coverage is measured over all eligible parcels and may fall below the nominal level in a finite realization; such a result is reported and never used to recalibrate the mass.

Taxonomic back-off uses a predeclared fourth-edition HCAT cut over the leaf catalogue. A confident parcel keeps its singleton; an uncertain parcel receives every represented leaf below the node containing the posterior maximizer. The committed mapping resolves every leaf and cut node against the EuroCrops taxonomy before constructing a prediction, and rejects missing, repeated, non-ancestral, or out-of-catalogue assignments. Three cut nodes contain only one represented study leaf; back-off in those cases leaves the visible singleton unchanged rather than jumping post hoc to a broader node. Consequently, a malformed hierarchy cannot silently turn a parcel into non-delivery, and a deeper hierarchy would be a different operating rule rather than a choice made after observing outcomes.

All operating points—the retained catalogue, confidence threshold, calibrated mass, and coarse partition—are inputs to these operators. None is fitted by them, and none may be re-equalized on the held-out block. This separation is what makes test-label invariance inspectable: changing the test labels cannot change a delivered set.

Invariance to the held-out labels is a property of the operators; disjointness of the blocks is a property of whoever calls them, and the implementation no longer assumes it. A fitted operating point carries the stable parcel identities of the rows used to fit it and refuses to be applied to any of them. Identity is not inferred from posterior bytes: distinct parcels may have identical posteriors, and recomputing the same parcel may change its bytes. Handing the evaluated block back as calibration data therefore fails rather than passing quietly, which is how the corresponding leak entered an earlier version of this work.

The implementation makes this temporal order part of the interface. Its fitting function accepts training and validation arrays but has no held-out argument; its application function accepts held-out posteriors but has no label argument; only the evaluator receives labels, after all six membership matrices exist. A single orchestration path then composes these stages without supplying a default catalogue size, delivery target, miscoverage level, back-off rate, hierarchy, or loss point. This prevents an absent preregistration choice from becoming a library default.

The leaf catalogue is fixed by the training-side model output. Separately, a neutral evaluation universe is selected once from training labels using the predeclared support floor. Validation and held-out labels cannot add or remove a class from that universe. Every arm retains every held-out parcel in its outcome denominator, including non-deliveries and parcels whose true class lies outside the neutral classwise summary. A selected class absent from a held-out block remains visible with zero support and an undefined classwise statistic; it is not silently dropped or converted to a valid zero. Thus every primary frequency and classwise diagnostic uses denominators fixed before the arm's delivery is observed.

4.2 The Complete Action–Outcome Grid

The outcome interface follows the full grid fixed by the loss-space contract. A precise label is either correct, the zero-loss reference, or wrong. An empty set is non-response. A multi-label set either contains or excludes the truth. A coarse class likewise either contains or excludes the truth. These are seven cells in total. The two failure cells for non-singleton answers are indispensable: without them, a prediction set or a coarse answer would be charged only when it succeeds and would again acquire a free failure mode.

Classification into a cell follows the delivery visible to the user, not merely the name of the algorithm. A singleton emitted by the set-valued arm is scored as a precise label. A multi-label delivery is distinguished as an unconstrained label set or a named coarse class, because the two may have different consequences.

An empty row is non-response under every arm. Every parcel maps to exactly one cell.

The primary evaluator counts all seven cells and returns their complete frequency vector without accepting a loss point. A separate sensitivity interface accepts six non-negative finite losses, requires the correct precise label to remain the zero reference, and refuses an incomplete mapping. After removing common scale, these points form a five-dimensional simplex. Synthetic points test the necessary property that changing the relative price of a wrong label and silence can reverse which action has lower risk; they do not become empirical preferences.

Finally, the evaluator divides every cell count by the same number of eligible parcels, exactly as in Eq. (2). Non-deliveries therefore contribute their count to the same denominator as every other outcome. Risk at any declared sensitivity point is the dot product in Eq. (3). Expected set size is reported separately as a descriptor. The earlier size-based utility remains useful for a diagnostic link to macro recall, but it is not the common currency used to rank the six arms.

5 Evaluation Protocol

The protocol first estimates each arm's complete outcome-frequency vector, then derives dominance and switching geometry over every normalized loss. It needs no point loss, and no empirical contrast was run to write this section.

5.1 Split Order and Pair Formation

The empirical banks are the optical modality of PASTIS-R [18] and BreizhCrops [17]. Naming PASTIS-R identifies the distributed bank; no radar channel enters the evaluated predictors.

Training fixes the neutral class universe and support-based catalogue, validation fixes the other operating-point components, and evaluation starts only after both are immutable. Stable parcel identities enforce disjoint splits and prevent applying a fitted point to any training or validation identity, including a copied row; posterior equality is irrelevant.

The common catalogue size is the smallest prefix of classes ordered by training support whose cumulative training mass reaches 0.9, with class id breaking support ties; the F-score and support rules use that same size. Validation targets a 0.9 delivery rate for abstention, a 0.1 miscoverage level for the adaptive set, and a 0.9 precise-leaf rate for taxonomic backoff. A class enters the neutral universe with at least 20 training parcels. These are author-chosen nominal intervention points, not equal-loss points, and none may be rematched on the held-out split.

The evaluator assigns every eligible held-out parcel under every arm to one of the seven closed outcomes. Pairing is by identity, with matching block and `patch_id`, so row reordering is allowed but omission, duplication, and substitution are not. The training-derived class universe travels with the profile and is never reconstructed from evaluation or a bootstrap draw.

Artifacts carry their evidential regime rather than inheriting it from filenames. A closed inventory covers the result root; each empirical record and custody row repeats one regime. Only sealed PDF or SVG design diagnostics directly under the figure directory are non-empirical exceptions. A confirmatory result must identify a frozen held-out dataset and descend from its preregistration; the current inventory is exploratory-only and cannot relabel either already-seen bank.

The third-dataset register contains 14 candidates from two screening rounds, none admissible. With no versioned query log, this supports only a bounded negative result over those entries, not an exhaustive search. Its label-cardinality, geography, spatial-split, and download criteria are author choices. The empty branch leaves both banks exploratory; reopening requires an amendment committed before reviewing another candidate.

5.2 Two Declared Uncertainty Views

For each arm pair and cost cell, the contrast is the left-arm indicator minus its comparator. The primary view resamples whole `patch_id` groups within blocks, carries every parcel and arm together, and combines blocks using their original parcel shares. There is no parcel bootstrap.

Separately, indicator differences are averaged within spatial blocks and those means receive equal weight. A paired Student interval over a declared coordinate or section exposes between-block variation without treating overlapping folds as independent regions. Its centre may differ from the parcel-population contrast when block sizes differ; forcing equality would change the estimand.

Both views retain the complete denominator. Below three unique paired units, the implementation returns deltas and a reason but no interval, p-value, or Holm eligibility. A comparison declared identical by construction is retained as a self-check and excluded from multiplicity.

5.3 Spatial Partition Selection and Sensitivity

Spatial-block selection precedes arm evaluation and sees only minimum test-to-training centroid distance and the construction buffer; the former must be one order of magnitude larger. This author-chosen threshold neither proves spatial independence nor diagnoses residual autocorrelation. The sealed profile uniquely selects five blocks; zero or multiple eligible candidates stop the run.

A retired rule required minimum class support in the worst block. It belonged to a classwise macro F-score, not the current mean loss, and remains a composition diagnostic rather than a selector.

After selection, every declared k is applied to the same paired indicators and the full curve is reported. Fingerprints ignore row order and block names; duplicate or changed k values, buffers, effective block counts, or parcel universes are errors. Finer partitions expose grouping sensitivity, never additional evidence or statistical power.

5.4 Power Has the Same Unit as Its Test

The minimum detectable effect for the paired block analysis is obtained from the noncentral Student distribution, not by adding two central-t quantiles. For n independent paired blocks, degrees of freedom $\nu = n - 1$, assumed standard deviation σ of the paired block differences, and test-wise level α^*, let

$$c = t_{1-\alpha^*/2,\nu}, \qquad \lambda = \frac{\delta\sqrt{n}}{\sigma}. \tag{4}$$

The implementation solves for the smallest absolute δ such that

$$\Pr\{T_\nu(\lambda) < -c\} + \Pr\{T_\nu(\lambda) > c\} \geq 1 - \beta. \tag{5}$$

The planning scale has an outcome-independent source and is never estimated from the assessed contrast; hypothetical unit counts use a predeclared effect, never the observed delta. Repeated identifiers cannot increase n, fewer than three units have no publishable MDE, and zero observed variance is not a zero planning scale. Unadjusted comparisons use $\alpha^* = \alpha$; a planned Holm family of size m uses the prospective first-step threshold $\alpha^* = \alpha/m$.

This calculation belongs only to the equal-block sensitivity estimand above. It does not describe the primary patch-cluster analysis, whose power depends on prospective outcome profiles and the cluster structure and is evaluated by simulation. Calling the block MDE the power of the primary estimand would recreate the unit mismatch that the two-view protocol was introduced to prevent. The code is executable on synthetic inputs, but no empirical MDE is reported here.

5.5 Dominance, Boundaries, and Multiplicity

Every stratum contains all unordered arm pairs and all six cost-bearing outcome coordinates. The primary scalar for an ordered pair is

$$D_{m,n} = \max_{j=1,\ldots,6} \left(q_{m,j} - q_{n,j}\right). \tag{6}$$

Observed componentwise dominance is descriptive. An inferential dominance claim requires a simultaneous one-sided upper bound for $D_{m,n}$ below zero, obtained from a maximum statistic over the complete predeclared family. Pairs or coordinates are not removed after their estimates are seen. If a required comparison is missing or has too few paired clusters, the affected simultaneous claim is withheld and the observed vectors remain available as descriptive output.

The frozen primary procedure uses a family-wise alpha of 0.05, 95% intervals, 9,999 paired patch-cluster resamples within spatial block, add-one bootstrap probabilities, and seed 20260909. Within each confirmatory stratum, the critical value is the maximum absolute studentized centred deviation over all 15 unordered arm pairs and 6 cost coordinates: 90 coordinates, or 180 directional inequalities. A zero standard error withholds the inferential claim, except where neither action can produce that outcome: their difference is exactly zero in every cluster, so that bound is the exact zero and carries no bootstrap probability. This structural bound is load-bearing: without it that pair cannot retain six bounded coordinates. Studentization is performed inside every resample with that resample's own block-centred patch-cluster standard error; reusing one denominator for all resamples is prohibited. A relation stated across predictors uses an intersection–union rule and must hold in the same direction in every predeclared predictor stratum; no favourable predictor is selected.

The fitted winner region of arm m is the intersection of the simplex with the half-spaces $(\boldsymbol{q}_m - \boldsymbol{q}_n)^\top \boldsymbol{\ell} \leq 0$ for every comparator n. The boundary for one pair is the corresponding equality. These objects use all coordinates and do not require a probability measure over the simplex. A plotted two-dimensional section must declare the coordinates it fixes; it cannot be labelled as the full map or converted into a universal scalar threshold.

The separate class-distribution analysis uses the class-unweighted mean absolute deviation of promise retention. For a catalogue arm, class retention is one exactly when the true leaf belongs to its training-fitted legend; for abstention it is the class-conditional rate of nonempty delivery. The two planned contrasts are F-score catalogue minus abstention and support catalogue minus abstention, with positive direction and one Holm family of size two. If any class in the training-fixed common universe is absent, the whole H2 family is withheld rather than changing the denominator. Observed equality does not establish equivalence.

Point losses are limited to non-confirmatory sensitivity cases with an external source or declared geometric section. They cannot select the operating point, define the primary family, or support a stakeholder claim. Earlier ratios and Fieller sets remain outside the primary analysis.

The target power is 0.8. Its sealed prospective alternative moves one percentage point out of each of the six adverse cells into the zero-loss reference and rotates the advantaged position through all six arms. Candidate admission uses 999 simulated datasets on the planned patch and block skeleton, the same seed and primary bootstrap, and intracluster correlations 0, 0.05, 0.1, and 0.2. The minimum one-sided 95% Wilson lower bound over arm placements and the correlation grid must reach the target. This author-chosen alternative is neither an observed effect nor a practical-importance margin. These values were frozen before empirical execution; this section introduces no result beyond the exploratory diagnostics reported next.

6 Results

We report exploratory full outcome accounting and predeclared diagnostics, not a universal preference between renunciation mechanisms. For the 5 frozen PASTIS-R predictors, none of the 75 within-predictor action pairs was certified as dominant: 71 were certainly incomparable because their simultaneous profiles crossed, while 4 remained undetermined. The latter are not ties and were not converted into evidence of equality.

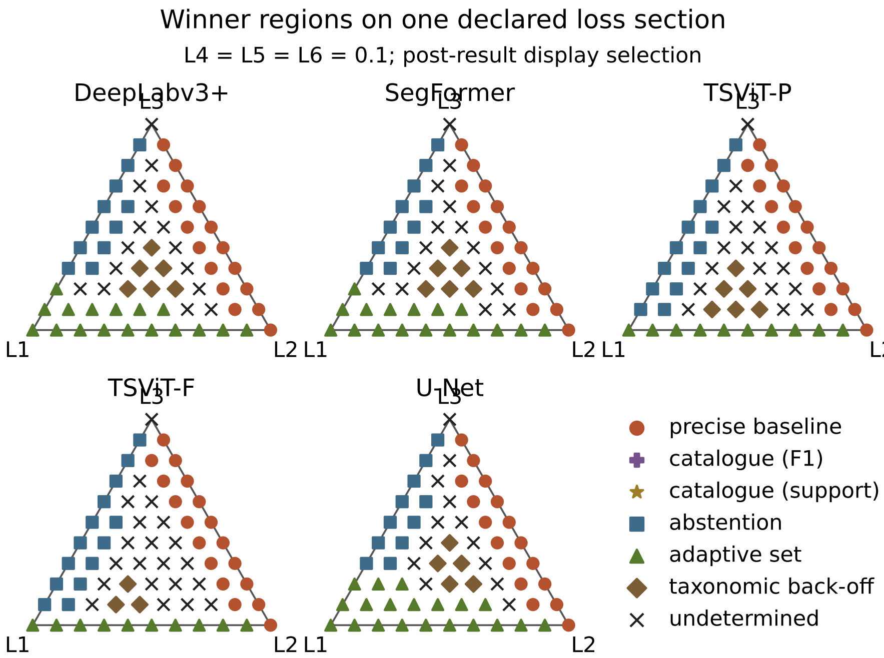

Fig. 1. Certain winner regions on one declared section of the exploratory PASTIS-R loss atlas. Symbols identify the winning action; crosses are evaluations for which the simultaneous band does not establish one winner. Each panel is a frozen predictor. Catalogue-restriction actions remain in the legend even though neither wins on this section, closing the six-action account. The displayed section is the first in the frozen enumeration and was selected for presentation after results.

The 20 declared two-dimensional sections contain 66 lattice points each. Across predictors this gives 6600 model–loss evaluations: 3913 had one certain winner and 2687 remained undetermined. The precise baseline won 2029 evaluations, adaptive sets 947, abstention 660, and taxonomic back-off 277; both catalogue-restriction rules won 0. These are counts on a declared finite grid, not probabilities, ranks, or fractions of the five-dimensional simplex. They therefore demonstrate switching, not how often an action should win under an undeclared distribution of losses.

Figure 1 shows the first section of the frozen exhaustive enumeration, with $L_4 = L_5 = L_6 = 0.1$ and no pooling across predictors. Its selection as the central display was made after the atlas was materialized and is disclosed rather than presented as preregistered. All 20 sections remain available in the structured artifact.

Retrospectively, an anonymized deployed system reduced eighteen classes to twelve using the largest top-F1 catalogue whose mean per-class F1 remained at least 0.90 on this same fold—not a per-class threshold. It emitted the restricted argmax only when the raw argmax was retained: 15265 of 16640 parcels (0.9174), comprising 13617 correct, 1648 wrong, and 1375 non-deliveries; four cells were structurally zero. The earlier 14688 (0.8827) counts reference labels inside the catalogue, not delivery coverage.

None of thirty frozen predictor–action configurations on these parcels dominated that profile; it dominated five and crossed twenty-five. These are complete-system comparisons, not an isolated mechanism effect or an identified implicit loss.

BreizhCrops does not contribute a paired inferential atlas. Each model–region stratum supplies 1 cluster under the frozen unit, below the minimum of 3, so no simultaneous interval, dominance claim, or phase boundary is estimable. Its outcome profiles remain descriptive; the dataset is not pooled with PASTIS-R to manufacture replication.

The conformal mass and the HCAT4 precise-delivery threshold were fitted on the frozen validation split. Both deliveries were fixed before evaluation labels were opened; reference availability only restricted the already-fixed PASTIS-R deliveries. Their validation targets are different operating constraints, not an equal loss.

Every conformal row met its marginal containment target, but only by returning multi-label answers. HCAT4 recovered additional truth-containing deliveries beyond correct precise answers in every row, yet also emitted wrong coarse answers in every row. Its realized precise-delivery rate varied across held-out strata despite one validation target. Thus a named coarse answer is neither a free rescue nor the same object as an arbitrary truth-containing set.

Rows use the complete bank-defined populations: 16640 valid-label PASTIS-R parcels and 30000 BreizhCrops parcels per external-region stratum. The other 11892 PASTIS-R posterior identities were excluded only after delivery was fixed, so file availability could not redefine the population.

Figure 2 changes the unit from parcels to mapped raster support while keeping the complete held-out denominator. Across every frozen predictor, adaptive sets allocated the smallest area share to deliveries guaranteed to exclude the truth, but moved most area into ambiguous truth-containing answers. Accordingly, every cropwise area interval under that arm had an unidentified direction. HCAT4 produced both identified signs and intervals spanning zero; singleton and non-delivery rules produced point errors because they leave no within-answer area allocation unresolved. None of these observations orders the arms without pricing ambiguity, error, and silence.

Here area means ten-metre reference-raster support inside each retained PASTIS-R parcel fragment, not cadastral or declared area. BreizhCrops contains neither geometry nor parcel area, so its surface view is not estimable and no proxy was substituted.

For every PASTIS-R predictor, nonempty-delivery spread across true classes exceeded that across training-defined area deciles for both catalogue rules and abstention: class support and parcel size are not interchangeable. BreizhCrops contributes class coverage only.

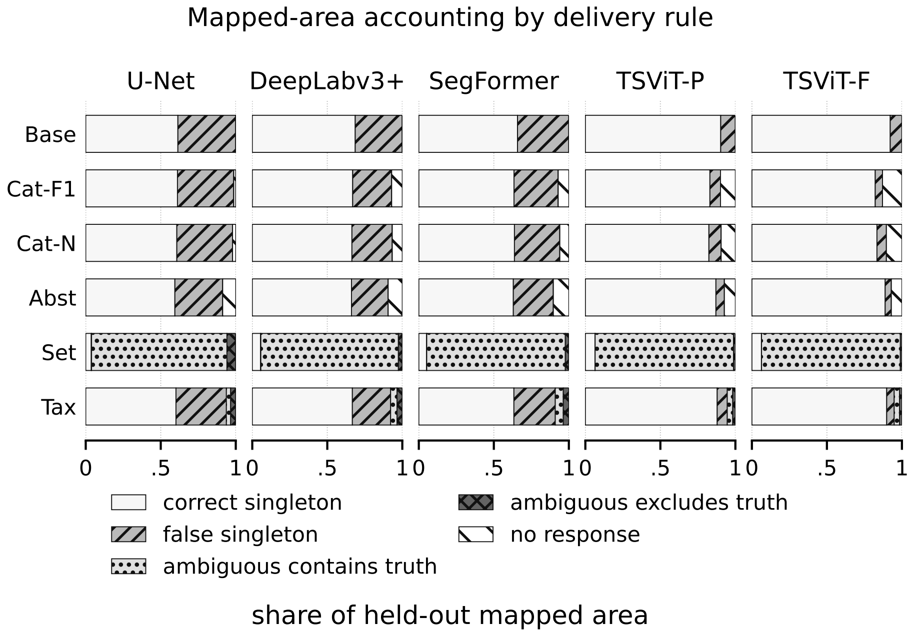

Fig. 2. Exploratory mapped-area accounting on the PASTIS-R held-out fold. Each bar partitions the complete evaluation area for one predictor and rule; predictors are not pooled. Catalogue, abstention, adaptive-set, and hierarchical rows retain distinct outcome categories. Fractions describe delivery accounting, not crop-area point estimates or utilities.

The frozen H2 family admitted no block interval. Only 2 of 5 PASTIS-R blocks contained all 18 training-declared classes, while each BreizhCrops model–region stratum supplied 1 block, below the minimum of 3. The predeclared rule withholds the complete family rather than changing the universe, inventing subregions, or pooling opposite-region fits; H2 therefore has no interval, p-value, or Holm decision.

These diagnostics explain the coordinates on which the atlas switches; they do not supply a loss point. Retired first-pass figures are not subtracted because their catalogue, predictor, and operating-point contexts differ.

7 Discussion

The diagnostics expose why one success rate cannot answer the decision question. Conformal sets attain marginal containment by returning multiple labels; HCAT4 sometimes recovers the truth through a communicable parent and sometimes returns the wrong parent. Calling either uniformly better would silently price breadth or coarse error at zero, so the outcome coordinates must remain separate.

The coverage statement is conditional on each evaluated bank and on exchangeability between its calibration and evaluation parcels. Both banks had been inspected before this execution, so every result here is exploratory even though the evaluation partitions are held out from calibration. No transport claim follows, and the closed dataset search did not find a separate admissible confirmatory benchmark. Predictor-to-predictor variation is sensitivity, not a route to select a favourable model. PASTIS-R therefore supports an exploratory decision atlas; BreizhCrops supports descriptive outcome profiles but not paired inference because its frozen reporting structure has only 1 cluster per model–region stratum.

Finding no certified dominance is not a null result. Most pairwise profiles certainly cross, and the lattice assigns certain wins to the precise baseline, abstention, adaptive sets, and taxonomic back-off in different loss regions. The data therefore reject a price-free global ranking while identifying where a reader's own loss vector changes the action. Catalogue restriction failing to win any sampled lattice point is narrower: it is not proof that its winning region has zero volume, because no measure over the simplex was declared and the finite grid cannot establish absence between its points.

No frozen alternative uniformly beats the deployed profile, so dismissing it requires a loss vector. But this is no within-predictor counterfactual: Full-M v2 lacks recorded training folds, the historical XGBoost member was not reproduced, and no separate calibration posterior survives. It audits what was served; it identifies neither a catalogue effect nor a held-out profile.

Mapped area makes the cost of ambiguity concrete without supplying its price. The adaptive set can avoid area that is guaranteed wrong by assigning that area to sets that usually contain the truth, yet those same sets leave every crop total only partially identified. HCAT4 narrows the admissible allocation through a named hierarchy, but a wrong parent is still a delivered error. An arm that looks safest under delivery validity can therefore be least informative for a crop-area total: a downstream consequence hidden by parcel accuracy, not a preference claim.

The unavailable H2 interval is a design result, not evidence that catalogue restriction and abstention distribute non-delivery equally. In PASTIS-R the frozen common-class requirement fails before either contrast is formed; in BreizhCrops the reporting unit supplies too few independent blocks. Both failures expose limits that a parcel count alone hides, and neither can be repaired by changing the denominator after observing class coverage.

Data and Code Availability

The anonymous submission archive contains the complete LaTeX project, bibliography, generated protocol appendix, and figures, but no raw satellite data. Machine-readable exploratory artifacts, execution contracts, and regenerating code remain private during double-blind review and will accompany a citable archival release before publication. Source data remain under their providers' access and licence terms.

A Frozen Analysis Contracts

Generated from sealed contracts, this appendix exposes frozen decisions but no observed loss, ranking, or contrast; missing or byte-divergent inputs stop the build.

Estimand and inferential units. Inference is conditional on each dataset and never pools datasets. The population is every eligible held-out parcel, including non-delivery. The analysis unit is the parcel; dependence is clustered by `patch_id`. The class universe is fixed from training, and the complete operating point is fixed from training and validation. Below 3 paired clusters or 3 paired spatial blocks, respectively, the corresponding view is descriptive: it has no interval, no p-value, and no Holm decision. No transport claim is made.

Complete loss space. Correct precise delivery is the zero-loss reference `R0`; the 6 cost-bearing cells are `L1`; `L2`; `L3`; `L4`; `L5`; `L6`. Removing only common positive scale produces a 5-dimensional simplex. The empirical object is each arm's outcome-frequency vector. Uniform dominance is componentwise; winner regions are exact linear half-spaces and pairwise switching boundaries are hyperplanes conditional on the remaining losses. No elicited point or affected-party preference is required or claimed.

Frozen inference. Family-wise alpha is 0.05; 9999 paired patch-cluster resamples use seed 20260909. The maximum covers 15 arm pairs by 6 costs (180 directional inequalities). Target power is 0.8 under 999 prospective simulations with a one-sided Wilson bound. The four operating targets are 0.9, 0.9, 0.1, and 0.9. H2 is one two-contrast Holm family. All are author choices.

Predictor panel. The predictor is a sensitivity factor, not a selected winner. Its 5 members from 4 families (minimum 3) are `unet`; `deeplabv3plus`; `segformer`; `tsvit-pheno`; `tsvit-pheno-fullm`. Any single-model selection must be nested outside evaluation; temporal status and family identities remain in the sealed contract.

Set-valued arm. The set-valued action uses deterministic, non-randomized split-conformal adaptive prediction sets. Calibration uses validation labels only. The score is cumulative probability through the true label; prediction returns the smallest nonempty probability-ranked prefix that reaches the calibrated mass, with stable class index as the tie-break. An unattainable finite-sample rank returns the complete catalogue containment over all held-out parcels.

Hierarchical arm. HCAT4 freezes every leaf-to-parent correspondence and fits delivery confidence on training and validation only (pastis-r: 18 leaves, 7 coarse actions, 3 approximate correspondences; breizhcrops-2017: 9 leaves, 4 coarse actions, 0 approximate correspondences). The sealed contract retains the complete named mapping.

Spatial partition. Selection uses a construction buffer of 1.0 km and requires a minimum centroid-separation ratio of 10.0. The rule is unique eligible candidate or stop; class support is diagnostic, not a selection input. The complete sensitivity grid is 5; 8; 10; 12; 15; 20; 25. These overlapping partitions are sensitivity views, not independent replications.

Class-imbalance sweep. The outcome-free grid contains 20 targets: catalogue sizes 3; 4; 5; 6 crossed with support exponents 0.0; 0.25; 0.5; 0.75; 1.0, at depth 1 and minimum support 100. Common support has 6 classes and the producer 6 closed outputs. This sweep is descriptive and unpooled.

Confirmatory-dataset search boundary. The versioned register contains 14 adjudicated candidate entries through 2026-09-08 and none admitted. With no preserved query log, this is bounded to those entries, not globally exhaustive. Eligibility is author-chosen; PASTIS-R and BreizhCrops remain exploratory. Reopening requires a prior versioned amendment.

Model identities. Frozen tuples are `tsvit-pheno`: $T = 10$, $d = 128$, train 1;2;3, validation 4, MD5 `b69582e3b55bb0fe8f4aa5d070a3a424`; `tsvit-pheno-fullm`: $T = 37$, $d = 192$, train 1;2;3, validation 4, MD5 `22ae8074b2c689565f8b854a56e4c5f5`; `tsvit-pheno-fullm-v2`: $T = 32$, $d = 192$, train not recorded, validation not recorded, MD5 `4a8d83d3fd0b593401725dbfbe2c3e22`. Missing folds remain 'not recorded' rather than inferred.

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